Blockchain Technology (CSE4444)
Assignment 2 Submission Document

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Assignment Title:	Blockchain Technology (CSE4444) - Assignment 2
Title of Developed DApp:	CertiTrust: Decentralized Academic Credential Verification System
GitHub Repository Link:	https://github.com/Aakarsh-Jha/Blockchain-Assignment-2

1. DApp Overview & Problem Statement

CertiTrust is an original decentralized application (DApp) designed to address academic credential fraud. Academic certificates and degrees are cryptographically registered by an authorized institution on the Ethereum blockchain. Each certificate record contains the student's name, degree title, document reference hash, and an immutable block timestamp. Employers, verification agencies, and students can instantly query the smart contract to verify authenticity.

2. Input Validation & Test Cases Summary

Test Case	Caller	Inputs	Observed Outcome
TC-01 (Valid)	Owner	Valid Name & Course	Mined & issued successfully (ID #1).
TC-02 (Invalid)	Owner	Empty student name ""	Reverted: 'Student name cannot be empty.'
TC-03 (Unauthorized)	Account 2	issueCertificate() call	Reverted: 'Unauthorized! Only authority can perform this action.'

3. Screenshots & Evidence

Figure 1: Smart Contract Compilation (Remix IDE)

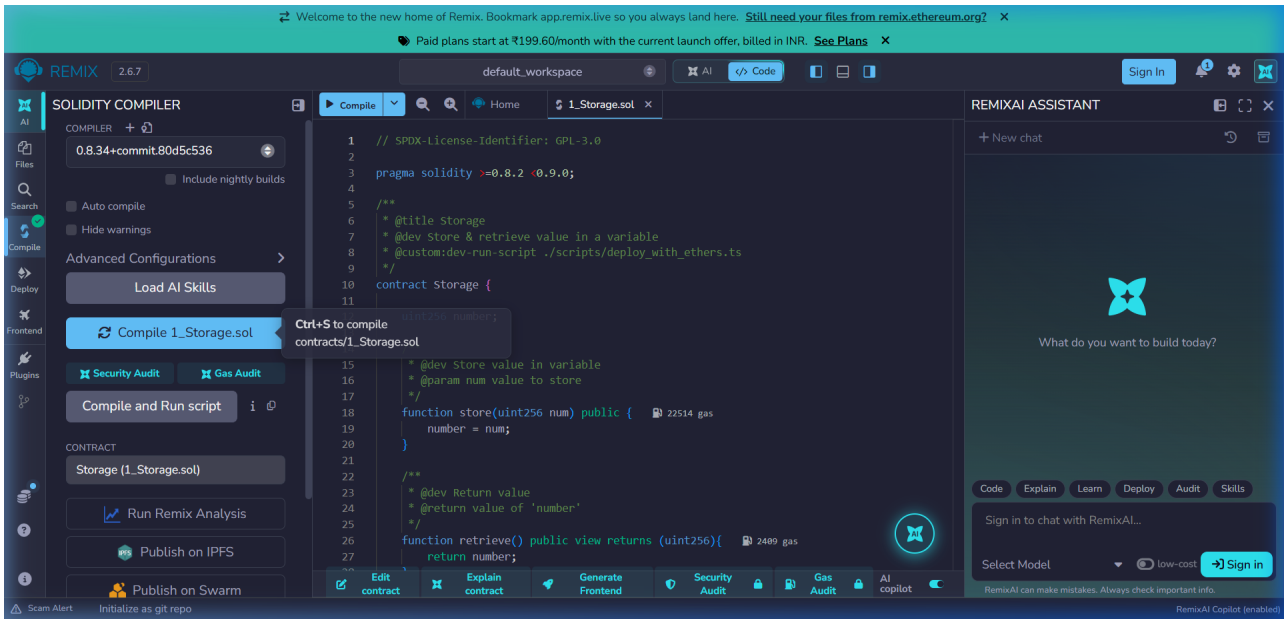


Figure 2: Smart Contract Deployment (Remix VM / Network)

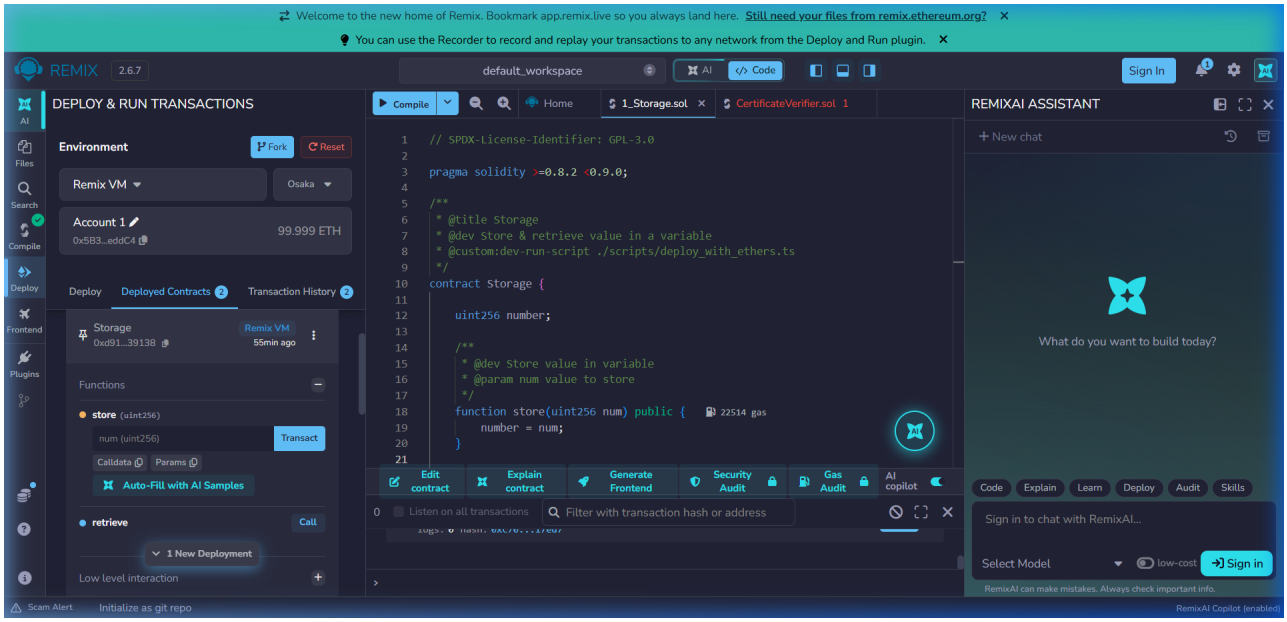


Figure 3: MetaMask Wallet Connection Handshake

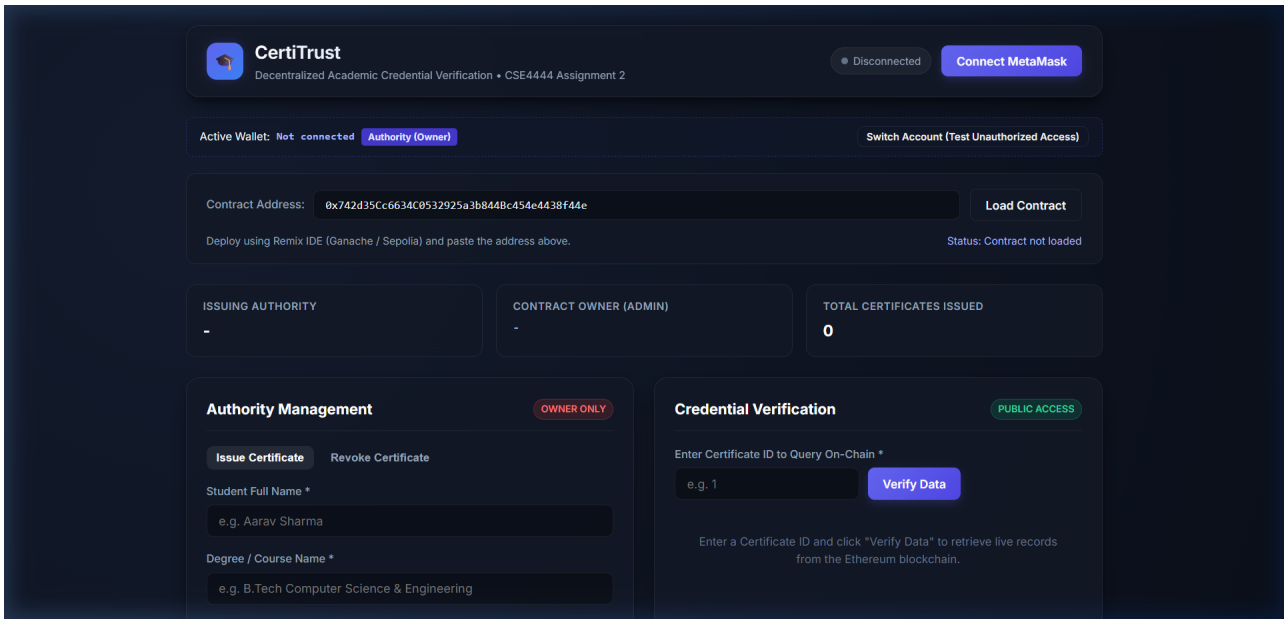


Figure 4: Successful Transaction Execution & Mining

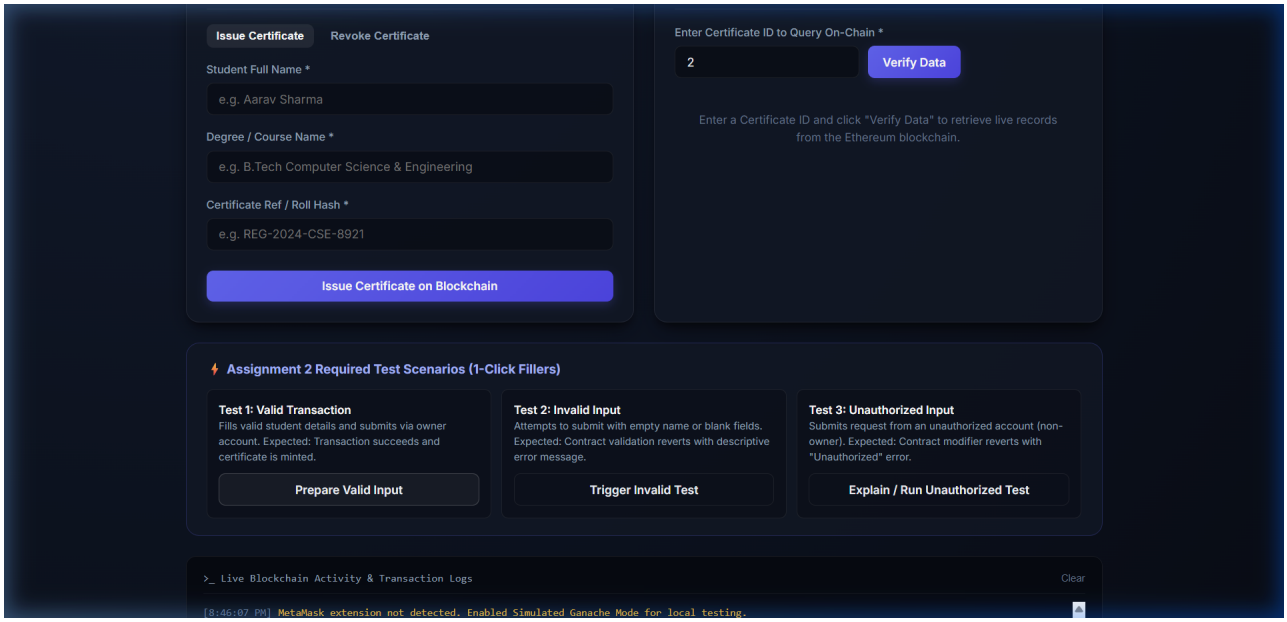


Figure 5: Failed Transaction Revert (Validation / Unauthorized)

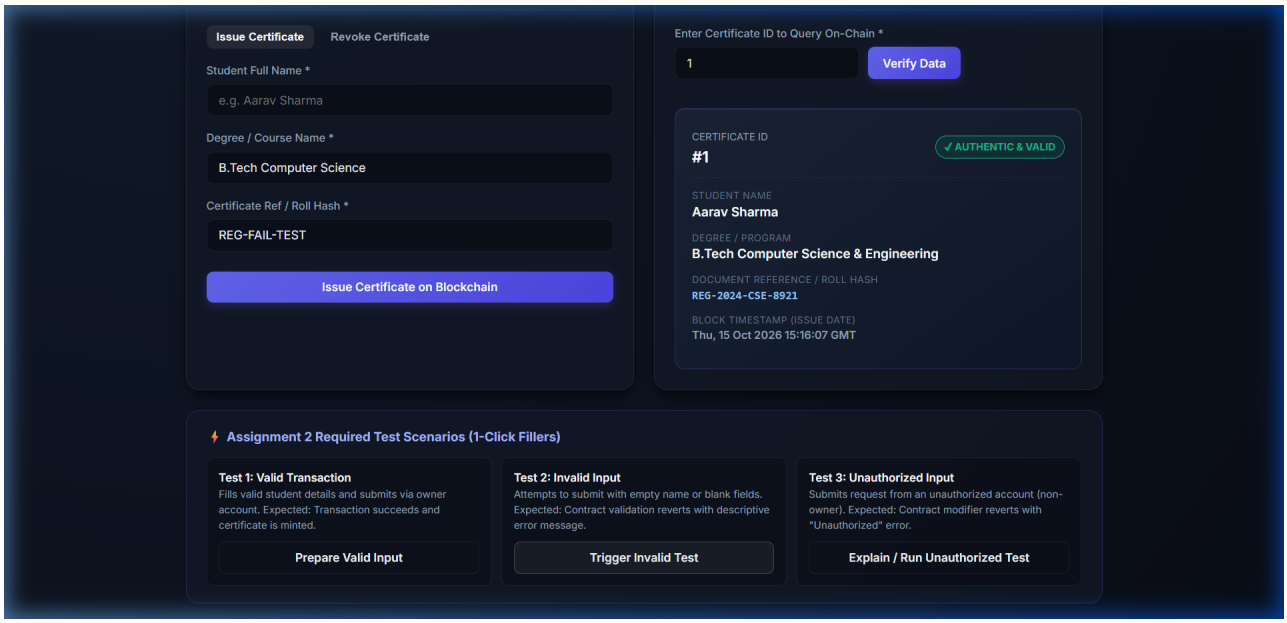


Figure 6: Blockchain Credential Retrieval & Verification in UI

